

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

QUIZ-2

DURATION: 40 MINUTES

SUMMER SEMESTER, 2023-2024

FULL MARKS: 15

Math 4641: Numerical Methods

Programmable calculators are not allowed. Write your Student ID at the top of your extra pages (if any).

Answer **all 4 (four)** questions. Figures in the right margin indicate full marks of questions whereas corresponding CO and PO are written within parentheses.

STUDENT ID:

1. One of the fundamental relationships between the secant and tangent functions is given by the following Pythagorean trigonometric identity, 7
(CO1)

$$\sec^2(x) - \tan^2(x) = 1 \quad (1) \quad \text{(PO1)}$$

One of the many proofs of this identity involves utilizing the Taylor expansions of its constituent transcendental functions.

The Taylor series for a function $f(x)$ is –

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + f^{(5)}(x)\frac{h^5}{5!} + \dots$$

Using the Taylor series (with at least the first 5 terms), derive the Maclaurin series of –

- i. $\sec^2(x)$
- ii. $\tan^2(x)$

Then use these two series to prove the Pythagorean trigonometric identity, as delineated in Equation 1.

Solution:

The Maclaurin series of a transcendental function is simply a Taylor series around the point $x = 0$.

- (i) Now, for $f(x) = \sec^2(x)$, let's calculate the derivatives at the point $x = 0$.

$$f(x) = \sec^2(x), f(0) = \sec^2(0) = 1;$$

$$f'(x) = 2\sec^2(x)\tan(x), f'(0) = 2\sec^2(0)\tan(0) = 0;$$

$$f''(x) = 4\sec^2(x)\tan^2(x) + 2\sec^4(x), f''(0) = 4\sec^2(0)\tan^2(0) + 2\sec^4(0) = 2;$$

$$f'''(x) = 8\sec^2(x)\tan^3(x) + 16\sec^4(x)\tan(x),$$

$$f'''(0) = 8\sec^2(0)\tan^3(0) + 16\sec^4(0)\tan(0) = 0;$$

$$f^{(4)}(x) = 16\sec^2(x)\tan^4(x) + 88\sec^4(x)\tan^2(x) + 16\sec^6(x), f^{(4)}(0) = 16;$$

Using the first 5 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + f^{(4)}(x)\frac{h^4}{4!} + \dots$$

For $x = 0$,

$$\begin{aligned}
 f(0+h) &= f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f''''(0)\frac{h^4}{4!} + \dots \\
 \Rightarrow f(h) &= 1 + 0 + \frac{2h^2}{2!} + 0 + \frac{16h^4}{4!} + \dots \\
 \Rightarrow f(x) &= 1 + 0 + \frac{2x^2}{2!} + 0 + \frac{16x^4}{4!} + \dots \\
 \Rightarrow \sec^2(x) &= 1 + 0 + \frac{2x^2}{2!} + 0 + \frac{16x^4}{4!} + \dots \\
 \Rightarrow \sec^2(x) &= 1 + x^2 + \frac{2x^4}{3} + \dots
 \end{aligned}$$

(ii) Now, for $f(x) = \tan^2(x)$, let's calculate the derivatives at the point $x = 0$.

$$\begin{aligned}
 f(x) &= \tan^2(x), f(0) = \tan^2(0) = 0; \\
 f'(x) &= 2 \tan(x) \sec^2(x), f'(0) = 2 \tan(0) \sec^2(0) = 0; \\
 f''(x) &= 2 \sec^4(x) + 4 \tan^2(x) \sec^2(x), f''(0) = 2 \sec^4(0) + 0 = 2; \\
 f'''(x) &= 16 \tan(x) \sec^4(x) + 8 \tan^3(x) \sec^2(x), f'''(0) = 0; \\
 f''''(x) &= 16 \sec^2(x) \tan^4(x) + 88 \sec^4(x) \tan^2(x) + 16 \sec^6(x), f''''(0) = 16;
 \end{aligned}$$

Using the first 4 terms of the Taylor series,

$$f(x+h) = f(x) + f'(x)\frac{h}{1!} + f''(x)\frac{h^2}{2!} + f'''(x)\frac{h^3}{3!} + \dots$$

For $x = 0$,

$$\begin{aligned}
 f(0+h) &= f(0) + f'(0)\frac{h}{1!} + f''(0)\frac{h^2}{2!} + f'''(0)\frac{h^3}{3!} + f''''(0)\frac{h^4}{4!} + \dots \\
 \Rightarrow f(h) &= 0 + 0 + \frac{2h^2}{2!} + 0 + \frac{16h^4}{4!} + \dots \\
 \Rightarrow f(x) &= 0 + 0 + \frac{2x^2}{2!} + 0 + \frac{16x^4}{4!} + \dots \\
 \Rightarrow \tan^2(x) &= \frac{2x^2}{2!} + \frac{16x^4}{4!} + \dots \\
 \Rightarrow \tan^2(x) &= x^2 + \frac{2x^4}{3} + \dots
 \end{aligned}$$

Now, to prove the Pythagorean trigonometric identity,

$$\begin{aligned}
 \text{L.H.S.} &= \sec^2(x) - \tan^2(x) \\
 &= 1 + x^2 + \frac{2x^4}{3} + \dots - \left(x^2 + \frac{2x^4}{3} + \dots \right) = 1 = \text{R.H.S.} \quad \square \text{ Q.E.D.}
 \end{aligned}$$

Rubric:

- 3 points for the correct series of $\sec^2(x)$.
- 3 points for the correct series of $\tan^2(x)$.
- 1 points for the correct proof.

2. Compare and contrast the following interpolation methods— **Direct** method, **Newton’s Divided Difference** method, and **Lagrangian** method. 1.5
(CO3)
(PO1)

Solution:

Any 2 valid comparisons will suffice. Some comparisons can be—

Criteria	Direct	Newton’s Divided Difference	Lagrangian
Interpolant Polynomial Representation	$f_n(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$	$f_n(x) = b_0 + b_1(x - x_0) + \dots + b_n(x - x_0)(x - x_1) \dots (x - x_{n-1})$	$f_n(x) = \sum_{i=0}^n \left(\prod_{j=0, j \neq i}^n \frac{x - x_j}{x_i - x_j} \right) f(x_i)$
Computational Complexity	Depends on the algorithm used to solve the equations. It may be $O(n^3)$, $O(n^3 \log(\ A\ + \ b\))$, or $O((n + 1)!)$.	Using memoized dynamic programming, we get a complexity of $O(n^2)$.	The naïve implementation is $O(n^2)$, but it can be optimized further down to $O(n \log^2(n))$.
Property of the terms	Each term has a coefficient a_i and these coefficients are obtained by solving all the equations simultaneously.	Each term has a coefficient b_i and each of these coefficients can be obtained by solving one equation.	The polynomial is a weighted sum of the given functional values $f(x_i)$, and each term is a product of the Lagrangian weight function $L_i(x)$ and the functional value $f(x_i)$.
Vulnerable to the Runge Phenomenon?	Yes	Yes	Yes

Rubric:

- 0.75 point for each valid comparison.

3. A thermistor, also known as thermal resistor, is a semiconductor type of resistor whose resistance is strongly dependent on temperature. To measure temperature using a thermistor, the manufacturers provide users with a Temperature (T) vs. Resistance (R) calibration curve. If we measure the resistance of the thermistor using an Ohmmeter, then we can refer to the aforementioned curve and determine the corresponding temperature value. Table 1 portrays multiple recorded observations involving a thermistor.

Table 1: Temperature T of a thermistor as a function of its resistance R for Question-3.

Resistance, R (Ω)	Temperature, T ($^{\circ}\text{C}$)
1101	25.113
911.3	30.131
636	40.12
451.1	50.128

Determine the value of the temperature T when the thermistor resistance is $R = 754.8\Omega$ using the Lagrangian method of interpolation and a second order polynomial.

Solution:

The 2nd order Lagrangian interpolant is –

$$T(R) = \sum_{i=0}^2 L_i(R)T(R_i)$$

$$= L_0(R)T(R_0) + L_1(R)T(R_1) + L_2(R)T(R_2)$$

Given, $R = 754.8\Omega$, and the 3 nearest data points that bracket it are, $R_0 = 911.3$, $R_1 = 636$, and $R_2 = 451.1$. The corresponding T values are $T(R_0) = 30.131$, $T(R_1) = 40.12$, and $T(R_2) = 50.128$ respectively.

Now, let's calculate the Lagrangian weights.

$$L_0(R) = \prod_{\substack{j=0 \\ j \neq 0}}^2 \frac{R - R_j}{R_0 - R_j} = \left(\frac{754.8 - 636}{911.3 - 636} \right) \times \left(\frac{754.8 - 451.1}{911.3 - 451.1} \right) = 0.284$$

$$L_1(R) = \prod_{\substack{j=0 \\ j \neq 1}}^2 \frac{R - R_j}{R_1 - R_j} = \left(\frac{754.8 - 911.3}{636 - 911.3} \right) \times \left(\frac{754.8 - 451.1}{636 - 451.1} \right) = 0.933$$

$$L_2(R) = \prod_{\substack{j=0 \\ j \neq 2}}^2 \frac{R - R_j}{R_2 - R_j} = \left(\frac{754.8 - 911.3}{451.1 - 911.3} \right) \times \left(\frac{754.8 - 636}{451.1 - 636} \right) = -0.218$$

Now that we have the weights, we can obtain the value of $T(754.8)$.

$$T(754.8) = 0.284 \times T(911.3) + 0.933 \times T(636) + (-0.218) \times T(451.1)$$

$$= 0.284 \times 30.131 + 0.933 \times 40.12 + (-0.218) \times 50.128$$

$$\approx 35.088^{\circ}\text{C}$$

Rubric:

- 1 point for the correct polynomial representation.
- 0.5 point for choosing the correct nearest points.
- 0.5 point for each Lagrangian weight $L_i(R)$ value.
- 1 point for the correct answer.

4. Prove that a polynomial of degree n or less that passes through $n + 1$ data points is unique.

2.5
(CO4)
(PO1)

Solution:

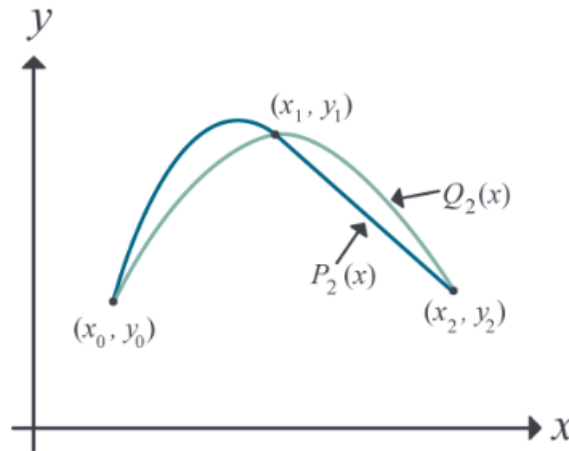


Figure 1: Interpolating polynomials through three points.

Proof: Let us use proof by contradiction. If the polynomial is not unique, then at least two polynomials of order n or less pass through the $n + 1$ data points.

Assume two polynomials $P_n(x)$ and $Q_n(x)$ go through $n + 1$ data points $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$. Then,

$$R_n(x) = P_n(x) - Q_n(x)$$

Since $P_n(x)$ and $Q_n(x)$ pass through all the $n + 1$ data points, we can say,

$$P_n(x_i) = Q_n(x_i) \quad \forall i \in \{0, \dots, n\}$$

Hence

$$\begin{aligned} R_n(x_i) &= P_n(x_i) - Q_n(x_i) \quad \forall i \in \{0, \dots, n\} \\ \Rightarrow R_n(x_i) &= 0 \end{aligned}$$

The n th order polynomial $R_n(x)$ has $n + 1$ zeros. A polynomial of order n can have $n + 1$ zeros only if it is identical to a zero polynomial, that is,

$$\begin{aligned} R_n(x) &\equiv 0 \\ \Rightarrow P_n(x) &\equiv Q_n(x) \end{aligned}$$

So, $P_n(x)$ and $Q_n(x)$ must be the same unique polynomial, which contradicts our initial assumption. $\square$ Q.E.D.

Rubric:

- 1 point for correctly defining $P_n(x)$, $Q_n(x)$, and $R_n(x)$.
- 1 point for showing $R_n(x)$ is a zero polynomial.
- 0.5 point for the final conclusion.